Apollo 3

An ontology-aware tool for manual curation of genomic feature annotations

Garrett J Stevens[1], Dario Beraldi[2], Shashank Gowda[3], William Haese-Hill[2], Colin Diesh[1], Matt Berriman[2], and Ian Holmes[1]

[1]Department of Bioengineering, University of California, Berkeley, CA, USA
[2]Wellcome Centre for Integrative Parasitology, Glasgow, UK
[3]EMBL-EBI, Hinxton, UK,

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Rich descriptions of feature types and structure

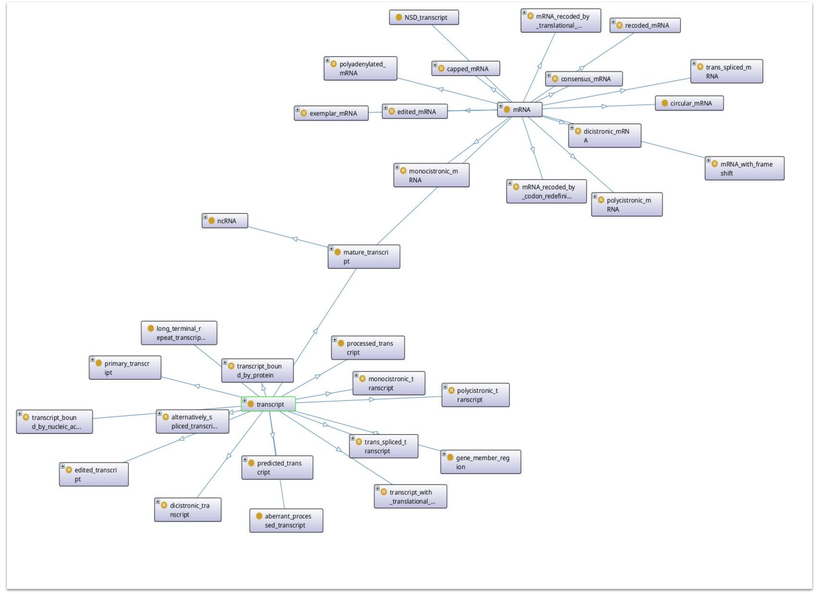

Fig. 1. Some of the many terms that are valid ways to describe a "transcript" (center of the bottom cluster) in the sequence ontology. Many databases represent transcripts with the term "mRNA," (center of the top cluster) without using any of its more descriptive sub-types.

The Sequence Ontology provides a rich set of terms used in biological sequence annotation. This ontology is often used to describe a genomic feature's "type" when using the GFF3 format.

Previous graphical manual annotation tools were limited, either in import from or export to GFF3, to a subset of the full Sequence Ontology.

Apollo 3 is designed to use an ontology (by default the Sequence Ontology) to support the import, export, and validation of the full range of terms, enabling more precise descriptions of genomic features. For example, Apollo 3 recognizes 204 different terms that can be used in place of "transcript" in a gene.

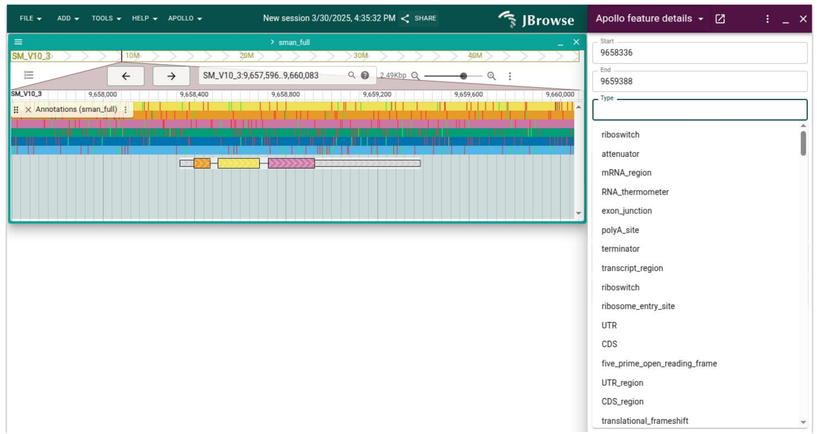

Fig. 2. When choosing a feature type, Apollo 3 pre-populates choices of all valid types based on the ontology, but also allows for custom entries. Apollo 3 supports

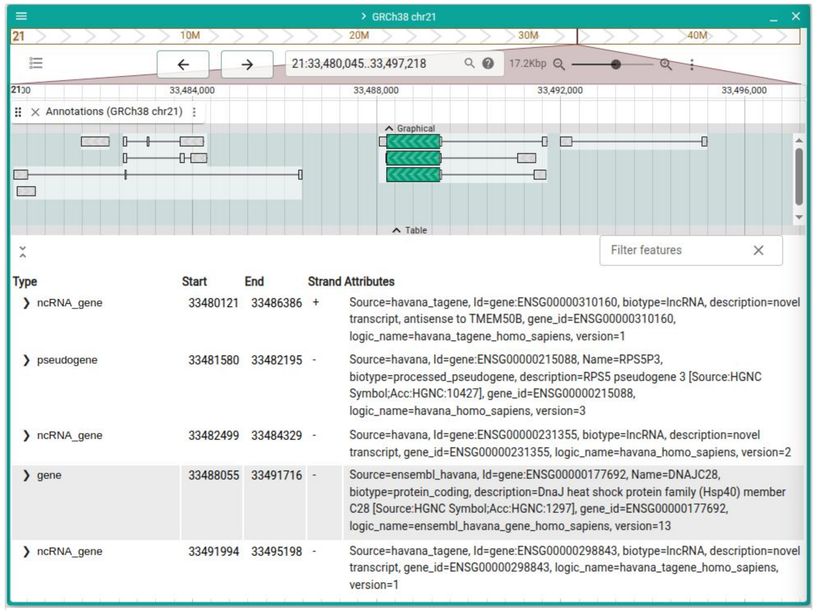

Fig. 3. In this region, "gene," "ncRNA_gene," and "pseudogene" are all recognized and rendered appropriately as "gene-like" features, based on Sequence Ontology types.

Functional annotation with any ontology

Beyond the Sequence Ontology, any ontology that can be described in the OBO Graphs JSON format can be loaded into Apollo. Once loaded, ontologies are searchable by ontology id, term, synonym, or description text, facilitating easy searching and assigning of relevant terms.

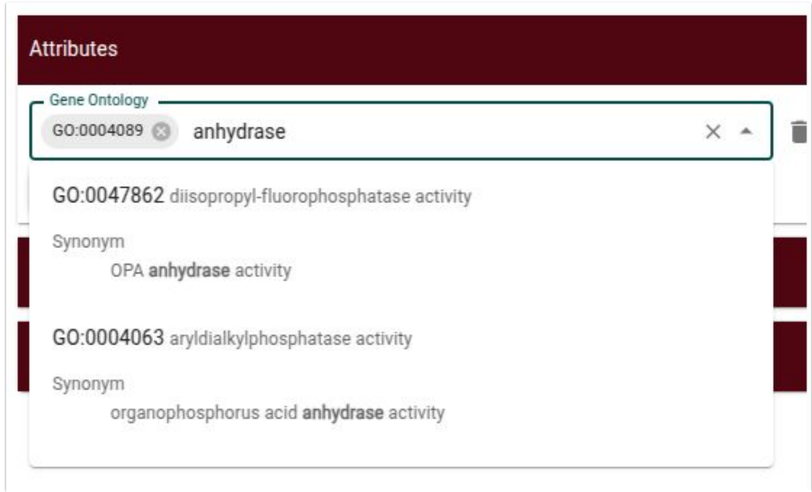

Fig. 4. A Gene Ontology functional annotation displaying the ability to search the ontology from within Apollo 3 and select multiple matching terms.